

Day – 28 : Joins in SQL

Types of SQL Joins

1. **INNER JOIN:** Retrieves records with matching values in both tables.
2. **LEFT JOIN (LEFT OUTER JOIN):** Retrieves all records from the left table and matching records from the right table. Non-matching records in the right table result in NULL.
3. **RIGHT JOIN (RIGHT OUTER JOIN):** Retrieves all records from the right table and matching records from the left table. Non-matching records in the left table result in NULL.
4. **FULL JOIN (FULL OUTER JOIN):** Retrieves all records from both tables, with NULL for non-matching rows in either table.
5. **CROSS JOIN:** Combines all rows from both tables, resulting in a Cartesian product.
6. **SELF JOIN:** Joins a table to itself.

To explain joins, we'll create two tables: **Employees3** and **Departments**.

Table 1: Employees

employee_id	first_name	last_name	department_id
1	Rahul	Sharma	101
2	Priya	Mehta	102
3	Ankit	Verma	103
4	Simran	Kaur	NULL
5	Aman	Singh	101

Table 2: Departments

department_id	department_name
101	Sales
102	Marketing
103	IT
104	HR

-- Create Employees3 Table

```
CREATE TABLE Employees3 (  
    employee_id SERIAL PRIMARY KEY,  
    first_name VARCHAR(50),  
    last_name VARCHAR(50),  
    department_id INT  
);
```

-- Insert Data into Employees3

```
INSERT INTO Employees3 (first_name, last_name, department_id)  
VALUES  
('Rahul', 'Sharma', 101),  
('Priya', 'Mehta', 102),  
('Ankit', 'Verma', 103),  
('Simran', 'Kaur', NULL),  
('Aman', 'Singh', 101);
```

-- Create Departments Table

```
CREATE TABLE Departments (  
    department_id INT PRIMARY KEY,  
    department_name VARCHAR(50)  
);
```

-- Insert Data into Departments

```
INSERT INTO Departments (department_id, department_name)  
VALUES  
(101, 'Sales'),  
(102, 'Marketing'),  
(103, 'IT'),  
(104, 'HR');
```

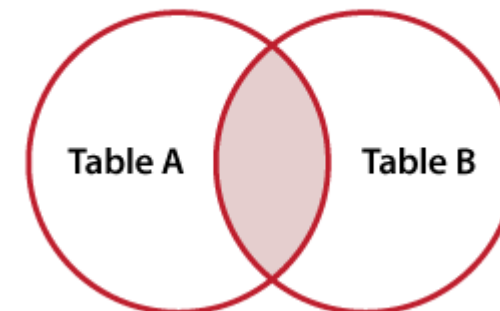
1. INNER JOIN - Retrieve Employees3 and their department names where a match exists.

```
SELECT
    e.employee_id,
    e.first_name,
    e.last_name,
    d.department_name
FROM
    Employees3 e
INNER JOIN
    Departments d
ON
    e.department_id = d.department_id;
```

OUTPUT →

employee_id	first_name	last_name	department_name
1	Rahul	Sharma	Sales
2	Priya	Mehta	Marketing
3	Ankit	Verma	IT
5	Aman	Singh	Sales

INNER JOIN



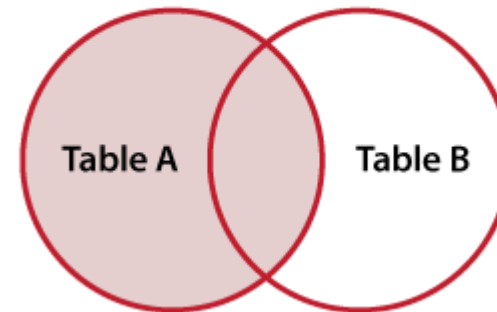
2. LEFT JOIN - Retrieve all Employees3 and their department names, including those without a department.

```
SELECT
    e.employee_id,
    e.first_name,
    e.last_name,
    d.department_name
FROM
    Employees3 e
LEFT JOIN
    Departments d
ON
    e.department_id = d.department_id;
```

OUTPUT →

employee_id	first_name	last_name	department_name
1	Rahul	Sharma	Sales
2	Priya	Mehta	Marketing
3	Ankit	Verma	IT
4	Simran	Kaur	NULL
5	Aman	Singh	Sales

LEFT OUTER JOIN



3. RIGHT JOIN - Retrieve all departments and the Employees3 working in them, including departments without

```
SELECT
    e.employee_id,
    e.first_name,
    e.last_name,
    d.department_name
FROM
    Employees3 e
RIGHT JOIN
    Departments d
ON
    e.department_id = d.department_id;
```

OUTPUT →

employee_id	first_name	last_name	department_name
1	Rahul	Sharma	Sales
2	Priya	Mehta	Marketing
3	Ankit	Verma	IT
NULL	NULL	NULL	HR

RIGHT OUTER JOIN

